



**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2018
YEAR SIX
H2 BIOLOGY (9744)**

Paper 3

Suggested Answers

Long Structured Questions

1a(i)

Process A: receptor-mediated endocytosis

Process B: acidification

(ii)

- Positive Single stranded RNA genome
- Directly use for translation after uncoating from nucleocapsid;

(iii)

- Formed/completed nucleocapsid enters rER
- Viral envelop is covered with pr proteins
- Virus is transport from rER to Golgi Apparatus via transport vesicle
- Furin protease modify/cleave pr protein in GA
- Virus is released from the cell

Max 4

b(i)

- seronegative refers to the absence of antibodies for any of the four serotypes of dengue virus
- seronegative children had never been infected by dengue virus

(ii)

- Considering that millions of children will eventually receive this vaccine
- AVP (e.g. 3 doses may be too expensive / hard to keep track for certain countries)

(iii)

- Antibody-dependent enhancement of infection occurs when pre-existing antibodies present in the body from primary infection, in this case, vaccination, bind to an infecting DENV particle of a different dengue serotype during subsequent infection
- Since vaccination results in immune system producing antibodies against all four serotypes, infection by any of the 4 serotypes will result in ADE
- During infection, heterotypic antibody binds to DENV
- Ab–virus complex attaches to Fcγ receptors (FcγR) on monocytes

1c(i)

Passive immunity

1c(ii)

- Antibodies from antivenom is complementary in shape to binding site of chlorotoxin

- Antibody is able to neutralise chlorotoxin by binding to binding site
- This prevents chlorotoxin from binding to / blocking Cl^- channels
- Cl^- channels can now allow Cl^- ions to pass in or out of the cell

1d

	Mithridatism	Antivenom
Type of immunity	Active	Passive
Type of injection	Venom containing low dosage of toxins (antigen)	Antibodies against toxins
Time taken to achieve immunity	Immune system takes a while to recognize antigen and produce antibodies	Immediate
Duration of immunity	Longlasting	Transient
Immunological memory	Generates immunological memory	Does not generate immunological memory

AVP;

2a

- most/ more years starting from 1998 shows an above average sea surface temperature as compared with past 100 years
- Peaks of high temperatures ranging from 0.5 to 1°C getting higher compared with past 100 years
- Starting from 1998, there have been higher numbers of countries around 10, reporting coral bleaching events compared to less than 10 before 1998
- There have also been higher number of countries reporting moderate to severe events in the last 18 years
- However, in 1998, there was an unusually high number of 74 countries reporting coral bleaching, which may not be directly due to high sea temperature since the temperature around that time was not exceedingly high

b

- Rising sea levels affecting the ability of zooxanthellae to photosynthesise due to lack of sunlight
- Increased carbon dioxide concentrations leading to acidification of oceans leading to less free carbonate ions are available for making calcium carbonate, a key component of the shells and skeletons of marine animals
- Extreme weather events like typhoons directly damaging corals

3a (i)

- Both consist of beta (1,4) glycosidic bonds
- Both consist of chains of polysaccharides lying parallel to one another
- 180° rotation of alternating/neighboring subunits

2 max**3a (ii)**

Features	Peptidoglycan	Cellulose
Cross-linking	Carbohydrate backbones crossed linked with short chains of amino acids/peptide cross bridges.	Cellulose molecule/chain (R: "cellulose") crossed linked with hydrogen bonds;
Repeating units	Consists of repeating units of NAM and NAG	Consists of repeating units of beta-glucose;

3a (iii)

- Hydrolysis of the glycosidic bonds
- Diffusion of water into the cells/high osmotic pressure ruptures the cell wall

3b (i)

Glycoprotein / glycolipid;

3b (ii)

- In intact ricin, the RTB polypeptide hides the catalytic site of RTA
- Only after RTA is separated from RTB (after the reduction process) does RTA exhibit its catalytic activity

3b (iii)

- Ricin is made as an inactive peptide
- Ricin is stored in the plant vacuoles, away from its substrates

3c (i)

- As temperature increases from 20 to 35 °C, the rate of reaction of free enzyme X increases from 0.60 to 0.95 A.U. due to the increased kinetic energy of the sugar substrate and enzyme molecules
- These molecules move faster colliding with one another in correct orientation to form more enzyme-substrate complexes
- The increased rate of reaction increases until it reaches its maximum rate of 0.95 A.U. at the optimum temperature of 35 °C when most enzyme-substrate complexes formed;
- Beyond the optimum temperature of 35 °C to 60 °C, rate of reaction decreases from 0.95 to 0.1 A.U. as intramolecular bonds are broken and three-dimensional conformation of active sites in enzymes is altered such that substrate can no longer fit

3 max**3c (ii)**

Description

- From 20 to 40 °C, activity of immobilised enzyme is lower than free enzymes (reference to any suitable data range) and the maximum activity of immobilised enzyme (0.80 A.U. at 35 °C) is lower than that of free enzyme (0.95 A.U. at 40°C/ between 40 to 45 °C);
1 max

Explanation

- Enzymes immobilised in alginate beads diffuse out from beads less easily to come into contact with sugar substrate/ less able to move and collide successfully with substrates leading to fewer enzyme-substrate complexes formed
- Products formed within the beads diffuse out less easily

1 max

Free Response Questions

4a Explain how the various aspects of DNA replication mechanism are important in the design of PCR technique and what additional considerations are needed for a PCR process. [10]

1. In replication, unwinding of DNA double helix by helicase to expose the parental template strands;
2. In the design of PCR, the parental template is exposed by heating till 95°C.;
3. Ref to breaking hydrogen bonds;
4. In replication, RNA primers are synthesized by primase;
5. In the design of PCR, forward and reverse DNA primers are added at 55°C;
6. Ref to formation of hydrogen bonds with template DNA;
7. In replication, DNA polymerase add complementary deoxyribonucleotides to 3'OH group of primer/ elongating strand;
8. In the design of PCR, Taq polymerase is added at 72°C;
9. Ref to formation of phosphodiester bonds;
10. Deoxyribonucleotide triphosphates (dNTPs) must be added to the PCR reaction;
11. This is because they are used as the monomers / building blocks during elongation;
12. Buffer solution must also be added to the PCR reaction;
13. To provide a suitable chemical environment for the reaction to occur + e.g. optimum pH / Magnesium chloride;
14. MgCl₂ is cofactor for DNA polymerase;

QWC: at least 2 aspects of DNA replication applied to design of PCR + at least one additional consideration discussed;

4b

Discuss the ethical implications of the application of stem cells in research and medical applications and how human induced pluripotent stem cells (iPSCs) overcome some of these issues. [15]

1. The sources of embryos in stem cell research are those generated for in vitro fertilization (IVF) that is no longer needed for reproductive purposes. Many potential donors have low fertility rates and a limited number of viable oocytes. Therefore, the prospect of reproductive success may be compromised because fewer oocytes are available for reproductive purposes.
2. As the procedures of oocyte retrieval are highly invasive and dangerous, the donor may be subject to medical risks such as ovarian hyperstimulation syndrome, bleeding, infection and complications of anaesthesia.
3. The blastocyst has the potential to develop into a person. Also, certain religious groups believe that life begins at fertilization. Therefore destroying the embryo is the same as killing a person.
4. The benefits of stem cell research come at the expense of human lives and rights. For people who believe that "human life begins at conception" and that an embryo is a person, an embryo would therefore have interests and rights that must be respected.
5. Slippery slope argument: Once we start down the path of the creation of life only to destroy it for other's purposes or benefits, then we will never be able to set an end to the dangers imposed on our "right to life." OR
[Medical application] Once we allow the use of technology for a group of people (patients), it would be unfair to deny other people the other benefits.
6. Since the donated embryos have no purpose and most will eventually be destroyed, they should be used for research instead of being wasted.
7. The blastocyst has the potential to develop into a person. However, it needs external aid to enable its development and hence it does not have an active potentiality to develop into a human being without help. In addition, certain religious groups believe that life begins at birth; hence destroying the embryo is not the same as killing a person.
8. If legalised, the focus of medical procedures of stem cells may be pushed towards other applications that are more profitable e.g. cosmetic surgery. This would cause the original good intentions of stem cell technologies to be lost.
9. Only the rich will be able to access the benefits of the expensive stem cell technology. The total cost for the procedure varies, but it can easily reach \$100,000 or more for an autologous transplant. Allogeneic transplants tend to cost even more, up to \$200,000 or higher.

10. hESC therapies has the potential to treat many diseases, including diabetes, Parkinson's disease, stroke, arthritis, multiple sclerosis, heart failure and spinal cord lesions. Since there are so many potential benefits to mankind, "the end justifies the means".
11. These procedures can only be performed by medical specialists and experts hence it is very safe and tightly regulated.
12. With advancement in technologies and competition, prices of novel medical procedures will eventually drop and become available for the masses. An example would be corrective eye surgery, Lasik, which has become cheaper and better with time.
13. iPSCs behave like ES cells; they can be injected into an early mouse blastocyst and produce all cell types within the developing embryo, and such embryos can complete gestation and are born alive.
14. This is achieved by using a retroviral expression vector to introduce four important stem cell factors into adult mouse somatic cells and reprogramming them into pluripotent stem cells.
15. The sources of cells in iPSCs are somatic cells which are easily and readily obtained without any lethal harm to the person.
16. Somatic cells used in iPSCs have no potential to become a person. Therefore there is no danger of destroying lives when iPSCs are used for research.
17. There is no removing of ICM and destroying the blastocyst. Hence using iPSCs does not involve destroying the embryo or killing a person.

14 Max.

+ 1 Mark for QWC

QWC- 3 marks each from arguments against and arguments for and 2 marks from iPSCs.

5a

5a Describe the mutation that give rise to sickle cell anaemia. [10]

- 1 Single base / nucleotide substitution within the gene;
- 2 Thymine substitutes with adenine within the gene coding for β -globin chain / sequence in gene is changed from CTT to CAT;
- 3 mRNA codon is changed from GAA to GUA;
- 4 Results in a change of amino acid, glutamic acid is replaced by valine;
- 5 Primary structure of β -globin chain is altered, leading to change in folding and coiling of the polypeptide;
- 6 Glutamic acid carries a negative charge & is polar / hydrophilic, while valine is non polar and hydrophobic;
- 7 Valine is located on the outside of the haemoglobin tetramer and interacts with a phenylalanine and a leucine from another haemoglobin molecule in order to get away from the water molecules;

- 8 This creates a hydrophobic spot on the outside of the protein structure that sticks to the hydrophobic region of an adjacent hemoglobin molecule's beta chain;
- 9 HbS molecule aggregates/ crystalises under low oxygen conditions, so is inefficient in carrying oxygen;
- 10 Polymerization of HbS molecules into rigid fibers causes red blood cells to change from circular biconcave shape to sickle shape;

9 Max.

+ 1 Mark for QWC (Showing central dogma of molecular bio, link from nucleotide to mRNA to protein)

5b Explain how 15 different strains of MRSA may arise from a single strain of *Staphylococcus aureus*. [15]

Methicillin Resistance

1. Mutation may have occurred in *S.aureus* by random chance, giving rise to methicillin resistance gene;
2. transformation, transduction, conjugation from other methicillin resistant bacteria species may have occurred which transfers plasmid with **methicillin resistant gene**;
3. Transformation occurs where there is uptake of foreign DNA coding for methicillin resistance by non-resistant *S.aureus*, **conferring methicillin resistance** as a result;
4. Transduction (general) occurs, where a bacteriophage accidentally incorporates DNA coding for methicillin resistance from MRSA and proceeds to infect non-resistant *S.aureus*, conferring methicillin resistance as a result;
5. Conjugation occurs, where F plasmid is transferred with methicillin resistant gene to non-resistant *S.aureus*, conferring methicillin resistance as a result;
6. Methicillin act as selection pressure;
7. Non-resistant bacteria are selected against / bacteria which are resistant to methicillin are selected for / they have a selective advantage;
8. MRSA survive, reproduce and pass down the gene coding for methicillin resistance to subsequent generations of bacterial cells (during binary fission);
9. Over many generations, frequency of gene coding for antibiotic resistance increases in the gene pool of bacteria;

Diversity

10. Mutation may have occurred in *S.aureus* by random chance, contributing to genetic diversity and multiple strains of *S.aureus*;
11. Horizontal gene transfer (transformation, transduction, conjugation) from other bacteria species may have occurred, **contributing to genetic diversity and multiple strains of *S.aureus***;
12. Transformation occurs where there is uptake of foreign DNA, contributing to genetic diversity and multiple strains of *S.aureus*;
13. Transduction (general) occurs, where a bacteriophage accidentally incorporates DNA from one bacteria strain and transfers it to another, contributing to genetic diversity and multiple strains of *S.aureus*;
14. Conjugation occurs, where F plasmid is transferred, contributing to genetic diversity and multiple strains of *S.aureus*;

QWC: Obtain at least 2 marks from both categories.